

MTH203: Sequences and series

Lecture 17

1. REARRANGEMENT AND ABSOLUTE CONVERGENCE

In the previous lecture, we saw that there exist rearrangements of the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ which converge to different quantities. The reason for this kind of behaviour is that this series is *conditionally convergent*. We will show that this sort of thing never happens for an *absolutely* convergent series. First, we note a rather simple lemma about convergence of sequences:

Lemma 1. *Let $(x_n)_{n \in \mathbb{N}}$ be a convergent sequence of real numbers, and let $x = \lim_{n \rightarrow \infty} x_n$. Let $(y_n)_{n \in \mathbb{N}}$. Suppose that for any $\epsilon > 0$, there exists an integer N such that for all $n > N$, we have $|x_n - y_n| < \epsilon$. Then $\lim_{n \rightarrow \infty} y_n = x$.*

Proof. Let $\epsilon > 0$ be given. There exists an integer N_1 such that for all $n > N_1$, $|x - x_n| < \epsilon/2$. Also, there exists an integer N_2 such that for all $n > N_2$, $|x_n - y_n| < \epsilon/2$.

Let $N = \max\{N_1, N_2\}$. Then, for any $n > N$, we have

$$|x - y_n| \leq |x - x_n| + |x_n - y_n| < \epsilon/2 + \epsilon/2 = \epsilon.$$

Thus, $\lim_{n \rightarrow \infty} y_n = x$. □

Exercise 2. Prove the converse of Lemma 1. In other words, prove that if $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ are two sequences of real numbers that converge to the same point, then for any $\epsilon > 0$, there exists N such that for all $n > N$, we have $|x_n - y_n| < \epsilon$.

We will also need the following lemma about absolutely convergent series:

Lemma 3. *Let $\sum_{n=1}^{\infty} x_n$ be an absolutely convergent series of real numbers. Then, for any $\epsilon > 0$, there exists an integer N such that if $A \subset \mathbb{N}$ is any finite subset such that $A \cap \{1, \dots, N\} = \emptyset$, then $|\sum_{k \in A} x_k| < \epsilon$.*

Proof. Let $\epsilon > 0$ be given. Since $\sum_{n=1}^{\infty} |x_n|$ is a convergent series, there exists an integer N such that for any integer $n \geq m > N$, $\sum_{k=m}^n |x_k| < \epsilon$.

Let $m = \min(A)$ and let $n = \max(A)$. Then, as $A \cap \{1, \dots, N\} = \emptyset$, we have $N < m \leq n$. Thus,

$$\left| \sum_{k \in A} x_k \right| \leq \sum_{k \in A} |x_k| \leq \sum_{k=m}^n |x_k| < \epsilon.$$

□

Exercise 4. Prove that the conclusion of Lemma 3 fails for conditionally convergent sequences. (For example, prove that it is not true for $\sum_{n=1}^{\infty} (-1)^{n+1}/n$.)

However, we have the following important result:

Theorem 5. *Let $\sum_{n=1}^{\infty} a_n$ be an absolutely convergent series of real numbers. Then, any rearrangement of the series converges to $\sum_{n=1}^{\infty} a_n$.*

Proof. Now, let $\sum_{n=1}^{\infty} b_n$ be a rearrangement of $\sum_{n=1}^{\infty} a_n$. Thus, there is a bijection $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ such that $b_n = a_{\sigma(n)}$. Let $\tau = \sigma^{-1}$, so that $b_{\tau(n)} = a_n$ for all n .

For any integer n , let $s_n = \sum_{k=1}^n a_k$ and $t_n = \sum_{k=1}^n b_k$. By Lemma 1, to show that the sequences $(s_n)_{n \in \mathbb{N}}$ and $(t_n)_{n \in \mathbb{N}}$ converge to the same point, it suffices to show that for any $\epsilon > 0$, there exists an integer M such that for any $n > M$, we have $|s_n - t_n| < \epsilon$.

Let $\epsilon > 0$ be given. The hypothesis says that the series $\sum_{n=1}^{\infty} |a_n|$ converges. Thus, by Lemma 3, for any $\epsilon > 0$, there exists an integer N such that for any finite subset $A \subset \mathbb{N}$ such that $A \cap \{1, \dots, N\} = \emptyset$, we have $\sum_{k \in A} |a_k| < \epsilon/2$. Let

$$M := \max\{\tau(1), \dots, \tau(N)\}.$$

Note that M is greater than or equal to the N distinct positive integers $\sigma(1), \dots, \sigma(N)$, and so $M \geq N$.

Let $n > M$. Let $S = \{\tau(1), \dots, \tau(N)\}$ and let $T = \{1, \dots, n\} \setminus S$. Then we have

$$\begin{aligned} t_n &= \sum_{k=1}^n b_k = \left(\sum_{k \in S} b_k \right) + \left(\sum_{k \in T} b_k \right) \\ &= \left(\sum_{k=1}^N b_{\tau(k)} \right) + \left(\sum_{k \in T} b_k \right) = \left(\sum_{k=1}^N a_k \right) + \left(\sum_{k \in T} b_k \right). \end{aligned}$$

We also have

$$s_n = \left(\sum_{k=1}^N a_k \right) + \left(\sum_{k=N+1}^n a_k \right).$$

Thus,

$$t_n - s_n = \left(\sum_{k \in T} b_k \right) - \left(\sum_{k=N+1}^n a_k \right)$$

and so

$$|t_n - s_n| \leq \left| \sum_{k \in T} b_k \right| + \left| \sum_{k=N+1}^n a_k \right|.$$

By the choice of N , we have

$$\left| \sum_{k=N+1}^n a_k \right| < \epsilon/2.$$

To obtain an estimate for the expression $\left| \sum_{k \in T} b_k \right|$, we note that if $k \in T$, then $k = \tau(n)$ for some $n > N$. In other words, $\sigma(k) > N$. Thus, $\sigma(T) \cap \{1, \dots, N\} = \emptyset$. So,

$$\left| \sum_{k \in T} b_k \right| = \left| \sum_{n \in \sigma(T)} b_{\tau(n)} \right| = \left| \sum_{n \in \sigma(T)} a_n \right| < \epsilon/2.$$

Using the last two inequalities, we get

$$|t_n - s_n| < \epsilon/2 + \epsilon/2 = \epsilon.$$

This proves the result. $\square$

We will come back to rearrangements in the next lecture.

2. EXTENDED REAL NUMBER SYSTEM

For some of our discussions below, it will be convenient to extend the real numbers by adding a couple of symbols.

We introduce the symbol ∞ (also written as $+\infty$) above we define it to be strictly larger than any real number. Similarly, we introduce the symbol $-\infty$ and define it to be strictly smaller than any real number. We also set the following conventions:

- (1) $\infty + \infty = \infty$.
- (2) $x + \infty = \infty$ for any $x \in \mathbb{R}$.
- (3) $x - \infty = -\infty$ for any $x \in \mathbb{R}$.
- (4) $x \cdot \infty = \infty$ for any $x > 0$.
- (5) $x \cdot \infty = -\infty$ for any $x < 0$.
- (6) $1/\infty = 1/(-\infty) = 0$.
- (7) $0 \cdot \infty$, $\infty - \infty$ and ∞/∞ are not defined.

Along with these operations, the set $\mathbb{R} \cup \{\infty, -\infty\}$ is called the *extended real number system*.

Definition 6. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence of real numbers.

- (a) We say that $\lim_{n \rightarrow \infty} x_n = \infty$ if for any $B \in \mathbb{R}$, there exists N such that for any $n > N$, we have $x_n > B$. In this case, we say that the sequence $(x_n)_{n \in \mathbb{N}}$ *diverges to ∞* .
- (b) We say that $\lim_{n \rightarrow \infty} x_n = -\infty$ if for any $B \in \mathbb{R}$, there exists N such that for any $n > N$, we have $x_n < B$. In this case, we say that the sequence $(x_n)_{n \in \mathbb{N}}$ *diverges to $-\infty$* .

Remark 7. If $(x_n)_{n \in \mathbb{N}}$ is a sequence such that $\lim_{n \rightarrow \infty} x_n = \infty$, or if $\lim_{n \rightarrow \infty} x_n = -\infty$, one can easily show that $\lim_{n \rightarrow \infty} 1/x_n = 0$. (Prove this as an exercise.) (Here we are assuming that $x_n \neq 0$ for all n so that it makes sense to talk about $1/x_n$.)

However, if $\lim_{n \rightarrow \infty} x_n = 0$, we cannot say that $\lim_{n \rightarrow \infty} 1/x_n = \infty$ or $-\infty$. Indeed, consider the sequence $((-1)^n/n)_{n \in \mathbb{N}}$. You can easily see that this sequence does not diverge to ∞ or $-\infty$.

However, if $(x_n)_{n \in \mathbb{N}}$ is a sequence such that $x_n > 0$ for all n , then if $\lim_{n \rightarrow \infty} x_n = 0$, it is certainly true that $\lim_{n \rightarrow \infty} 1/x_n = +\infty$. (Prove this as an exercise.)

3. POWER SERIES

A power series is a formal sum of the form $p(X) := \sum_{n=0}^{\infty} a_n X^n$ where X is a variable and the a_n 's are *constants*, i.e. numbers in some appropriate set. The a_n 's are called the *coefficients* of the series. If all the coefficients are real numbers, we call this a power series in X with *real coefficients*. We will only consider power series with real coefficients.

Thus, this is very much like a polynomial, except that it has terms of arbitrarily high degree in X . Just like a polynomial, one might be tempted to “substitute” a real number in place of X and thus view the power series as a function. For a real number t , we will denote the series $\sum_{n=0}^{\infty} a_n t^n$ by $p(t)$. If this series converges, we will say that the power series p *converges at t* . If it converges absolutely, we will say that p *converges absolutely at t* . If p does not converge at t , we say that p *diverges at t* .

Theorem 8. Let $p(x) = \sum_{n=0}^{\infty} a_n X^n$ be a power series in X . Suppose the series p converges at a real number $t \in \mathbb{R}$. Then p converges absolutely for any s such that $|s| < |t|$.

Proof. Since $\sum_{n=0}^{\infty} a_n t^n$ converges, the terms $a_n t^n$ are bounded. Thus, there exists a real number M such that $|a_n t^n| < M$. This number can now be used to estimate the terms of the series $p(s)$ for any real number s .

Suppose s is such that $|s| < |t|$. Then

$$|a_n s^n| = |a_n t^n| \cdot |(s/t)^n| \leq M \cdot |s/t|^n.$$

Since $|s/t| < 1$, the series $\sum_{n=0}^{\infty} M \cdot |s/t|^n$ converges. Thus, by the Comparison Test, the series $\sum_{n=0}^{\infty} |a_n s^n|$ also converges. In other words, p converges absolutely at s . $\square$

For a power series $p(X)$, let us define A to be the set of all non-negative real numbers x such that there exists $t \in \mathbb{R}$ such that $p(t)$ converges and $|t| = x$. If A is bounded above, we define $R = \sup(A)$. If A is not bounded above, we set $R = \infty$. We will call R as the *radius of convergence of p* .

Theorem 9. Let $p(X)$ be a power series and let R be its radius of convergence. If x is such that $|x| < R$, then p converges absolutely at x . Also, if x is such that $|x| > R$, then p diverges at x .

Proof. As we discussed above, let A be the set of all non-negative real numbers x such that there exists $t \in \mathbb{R}$ such that $p(t)$ converges and $|t| = x$. By definition, since $|x| < R$, $|x|$ is not an upper bound for A . Thus, there exists some $t \in \mathbb{R}$ such that p converges at t and $|x| < |t|$. By Theorem 8, p converges absolutely at x . $\square$

Thus, we see that if R is the radius of convergence of the power series $p(X)$, then p converges absolutely at every element of $V_R(0)$ (the ball of radius R with center 0, see Lecture 7). This explains the terminology. Note that, in general, we do not know whether p converges at $\pm R$. For example, the power series $\sum_{n=1}^{\infty} X^n/n$ converges at -1 , but diverges at 1 .

Theorem 10. Let $p(X) = \sum_{n=0}^{\infty} a_n X^n$ be a power series.

- (a) Suppose $\lim_{n \rightarrow \infty} |a_{n+1}/a_n| = L$ where $L \in \mathbb{R} \cup \{\infty\}$. Then the radius of convergence of p is $1/L$.
- (b) Suppose $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L$ where $L \in \mathbb{R} \cup \{\infty\}$. Then the radius of convergence of p is $1/L$.

Proof. We first prove part (a). Let $L = \lim_{n \rightarrow \infty} |a_{n+1}/a_n|$. We will use the Ratio Test. The limit of the ratio of successive terms of the series $\sum_{n=0}^{\infty} a_n t^n$ is

$$\lim_{n \rightarrow \infty} \frac{|a_{n+1} t^{n+1}|}{|a_n t^n|} = \lim_{n \rightarrow \infty} |t| \cdot \left| \frac{a_{n+1}}{a_n} \right| = |t| \cdot L.$$

Thus, we see that the series converges absolutely if $|t|L < 1$, i.e. $|t| < 1/L$. This proves part (a).

The proof of part (b) is similar. $\square$

Examples 11.

- (1) Let $p(X) = \sum_{n=0}^{\infty} X^n$. Then radius of convergence of this power series is easily seen to be 1. We have seen that for any real number t such that $|t| < 1$, we have $p(t) = \frac{1}{1-t}$.
- (2) Let $p(X) = \sum_{n=0}^{\infty} \frac{X^n}{n!}$. This is called the *exponential power series*. Since $\frac{1/(n+1)!}{1/n!} = \frac{1}{n+1}$, we see that the radius of convergence of this series is ∞ . The number $p(1) = \sum_{n=0}^{\infty} 1/n!$ is generally denoted by e .
- (3) Let $p(X) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \cdot X^n$. Then,

$$\lim_{n \rightarrow \infty} \frac{1/n + 1}{1/n} = \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1.$$

Thus, the radius of convergence of this series is 1. This power series is usually denoted by $\log(1+x)$.

4. READING SUGGESTIONS

This material in Section 1 is from Sections 2.1 and 2.7 of the textbook ([1]). The remaining material is from Chapter 6 of the book. (While the rest of Chapter 6 is not part of the syllabus, you are expected to know what has been discussed in this lecture.)

REFERENCES

- [1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015. [4](#)